

Equidistant-Step Euler-Type Methods for Time-Changed Stochastic Differential Equations

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Outline

- 1 Background and motivation
- 2 Strong convergence
 - Inverse subordinator estimates
 - Globally Lipschitz coefficients: EM
 - Super-linear coefficients: TEM
- 3 Long-time stability
 - Mean-square polynomial stability: EM
 - Almost sure stability: TEM
- 4 Numerical experiments
- 5 Summary

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Why time change?

Financial markets. In markets with infrequent transactions, asset prices may remain unchanged during long intervals without trades.

(Magdziarz et al., JSP, 2011)

Intracellular transport. Insulin granules alternate between correlated motion and random trapping inside pancreatic β -cells. (Tabei et al., PNAS, 2013)

Animal foraging. Raptors alternate between fast commuting and long, irregular waits during local search. (Vilk et al., PRX, 2022)



Classical versus time-changed

- **Classical Itô SDE**

$$\begin{aligned} dX(t) &= f(X(t))dt + g(X(t))dB(t) \\ \implies \frac{\partial p}{\partial t} &= -\frac{\partial}{\partial x}(f(x)p) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(g^2(x)p) \end{aligned}$$

- **Time-changed SDE** (inverse α -stable E)

$$\begin{aligned} dX(t) &= f(X(t))dE(t) + g(X(t))dB(E(t)) \\ \implies D_t^\alpha p &= -\frac{\partial}{\partial x}(f(x)p) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(g^2(x)p) \end{aligned}$$

Here D_t^α denotes the Caputo derivative of order $\alpha \in (0, 1)$.

Basic setting

Probability space. We work on a complete filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$.

Subordinator D a non-decreasing Lévy process with $D(0) = 0$, determined by its Laplace exponent ψ :

$$\mathbb{E}\left[e^{-sD(t)}\right] = e^{-t\psi(s)}, \quad \psi(s) = \vartheta s + \int_0^\infty (1 - e^{-sy})\nu(dy),$$

where $\vartheta \geq 0$ and the Lévy measure satisfies $\int_0^\infty (y \wedge 1)\nu(dy) < \infty$.

Inverse subordinator E

$$E(t) := \inf\{u > 0 : D(u) > t\}.$$

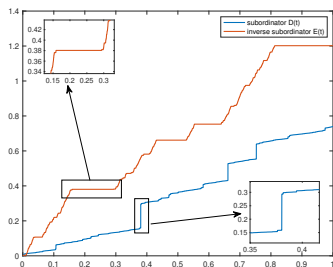
In the cases considered here, D has strictly increasing paths. Hence, E is continuous and non-decreasing.

In particular, we consider the α -stable case, where $\psi(s) = s^\alpha$, $\alpha \in (0, 1)$.

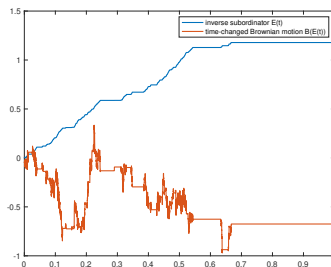
$$\mathbb{E}[E^n(t)] = \frac{n!t^{n\alpha}}{\Gamma(n\alpha + 1)}, \quad t \geq 0, \quad n \in \mathbb{N}.$$

Let B be an m -dimensional Brownian motion independent of D . Write $\mathbb{E}_B$ and $\mathbb{E}_D$ for the corresponding expectations.

Sample paths



D (blue) and its inverse E (red), $\alpha = 0.8$.



E (blue) and $B \circ E$ (red).

Previous numerical methods: existing works

$$dX(t) = h(t, E(t), X(t))dt + f(t, E(t), X(t))dE(t) + g(t, E(t), X(t))dB(E(t)).$$

✓ Euler–Maruyama method

- Jum E., Kobayashi K., PMS, 2016 ($h = 0$, $f(E, X)$, $g(E, X)$)
- Jin S., Kobayashi K., JMAA, 2019 ($h = 0$, $f(t, X)$, $g(t, X)$)
- Jin S., Kobayashi K., BIT, 2021 ($h(E)$, $f(E, X)$, $g(E, X)$)

✓ Semi-implicit and truncated Euler methods

- Deng C., Liu W., BIT, 2020 ($h = 0$, $f(E, X)$, $g(E, X)$)
- Liu W. et al., ANM, 2020 ($h = 0$, $f(E, X)$, $g(E, X)$)
- Li X. et al., JCM, 2025 ($h = 0$, $f(t, X)$, $g(t, X)$)

✓ Other schemes

- Wang S. et al., preprint, 2025 (semi-implicit and projected Euler, $h = 0$, $f(E, X)$, $g(E, X)$)
- Chen Z. et al., ANM, 2026 (stochastic θ -method, Lévy noise, $h = 0$, $f(E, X)$, $g(E, X)$)
- Wu R., Z., ANM, 2026 (Milstein-type, $h = 0$, $f(t, X)$, $g(t, X)$)

Carnaffan S., Kawai R., SISC, 2017 did use an **equidistant** grid, but gave no convergence analysis.

Earlier Lamperti result

Liu, Wang and Z., earlier work

We start from the scalar multiplicative-noise model

$$dY(t) = a(Y(t))dE(t) + b(Y(t))dB(E(t)).$$

The Lamperti transformation

$$X = F(Y), \quad F(y) = \lambda \int^y \frac{dz}{b(z)},$$

reduces it to the additive-noise equation

$$dX(t) = \lambda \left(\frac{a}{b} - \frac{1}{2} b' \right) (F^{-1}(X(t))) dE(t) + \lambda dB(E(t)).$$

On the equidistant physical-time grid $t_n = n\Delta t$, we then apply the backward Euler–Maruyama method

$$X_{n+1} = X_n + f(X_{n+1})\Delta E_n + \lambda\Delta B_n,$$

For suitable original models, this gives the parameter-related L^1 bound

$$\max_{0 \leq n \leq N} \mathbb{E}|X(t_n) - X_n| \leq C\Delta t^\alpha.$$

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Two estimates used in the convergence analysis

Let E be the inverse of an α -stable subordinator. The proof requires two complementary estimates.

Exponential moments. For $C > 0$,

$$\mathbb{E}[e^{CE(t)^p}] < \infty, \quad 0 < p < (1 - \alpha)^{-1}.$$

This controls the random factors produced by Gronwall's inequality.

Lemma 1 (largest increment of E)

For $h > 0$, define

$$\Xi_h := \sup_{0 \leq u \leq T-h} (E(u+h) - E(u)).$$

Then, for each $n \geq 1$ and $\varepsilon \in (0, \alpha)$,

$$\mathbb{E}[\Xi_h^n] \leq Ch^{n(\alpha-\varepsilon)}, \quad h \in (0, \tfrac{1}{2}].$$

The variable Ξ_h controls the local time-change error uniformly over the physical-time grid.

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Globally Lipschitz setting

For the time-changed SDE

$$dX(t) = f(X(t))dE(t) + g(X(t))dB(E(t)), \quad X(0) = X_0 \in \mathbb{R}^d,$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $g: \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$.

We first consider the following conditions on f and g .

Assumption 2 (global Lipschitz)

There is a constant $K > 0$ such that, for all $x, y \in \mathbb{R}^d$,

$$|f(x) - f(y)| + |g(x) - g(y)| \leq K|x - y|.$$

Assumption 3 (linear growth)

There is a constant $K > 0$ such that, for all $x \in \mathbb{R}^d$,

$$|f(x)| + |g(x)| \leq K(1 + |x|).$$

Equidistant-step EM

Fix the equidistant grid $t_n = n\Delta t$ and set

$$X_{n+1} = X_n + f(X_n)\Delta E_n + g(X_n)\Delta B_n,$$

where $\Delta E_n = E(t_{n+1}) - E(t_n)$ and $\Delta B_n = B(E(t_{n+1})) - B(E(t_n))$.

Two continuous-time versions

$$\tilde{X}_t = X_n, \quad t \in [t_n, t_{n+1}).$$

$$\bar{X}_t = X_0 + \int_0^t f(\tilde{X}_r) dE(r) + \int_0^t g(\tilde{X}_r) dB(E(r)).$$

They agree on the grid, $\bar{X}_{t_n} = \tilde{X}_{t_n} = X_n$.

EM moment bounds

Under Assumptions 2–3, the following estimates hold.

Proposition 4 (exact solution)

For every $p \geq 1$,

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X(t)|^p \right] < \infty.$$

Lemma 5 (EM interpolations)

For every $p \geq 2$,

$$\sup_{0 < \Delta t \leq 1} \mathbb{E} \left[\sup_{0 \leq t \leq T} (|\bar{X}_t|^p + |\tilde{X}_t|^p) \right] \leq C.$$

EM error estimates

Under Assumptions 2–3, let $\alpha \in (1/2, 1)$, $2 \leq p < (1 - \alpha)^{-1}$ and $\varepsilon \in (0, \alpha)$.

Theorem 6 (step-process error)

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X(t) - \tilde{X}_t|^p \right] \leq C \Delta t^{\frac{p(\alpha - \varepsilon)}{2}}.$$

Lemma 7 (interpolation gap)

For all sufficiently small $\Delta t > 0$,

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |\bar{X}_t - \tilde{X}_t|^p \right] \leq C \Delta t^{\frac{p(\alpha - \varepsilon)}{2}}.$$

EM convergence

Theorem 8 (main result)

Let Assumptions 2–3 hold, let $\alpha \in (1/2, 1)$, $2 \leq p < (1 - \alpha)^{-1}$ and $\varepsilon \in (0, \alpha)$. Then there is $C > 0$, independent of Δt , such that, for all sufficiently small $\Delta t > 0$,

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X(t) - \bar{X}_t|^p \right] \leq C \Delta t^{\frac{p(\alpha - \varepsilon)}{2}}.$$

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Khasminskii-type assumptions

Assumption 9 (polynomially weighted local Lipschitz)

There are constants $L > 0$ and $\gamma > 0$ such that, for all $x, y \in \mathbb{R}^d$,

$$|f(x) - f(y)| \vee |g(x) - g(y)| \leq L(1 + |x|^\gamma + |y|^\gamma)|x - y|.$$

This implies $|f(x)| \vee |g(x)| \leq M(1 + |x|^{\gamma+1})$ for some $M > 0$.

Assumption 10 (monotonicity)

There are constants $p > 2$ and $K > 0$ such that, for all $x, y \in \mathbb{R}^d$,

$$(x - y)^\top (f(x) - f(y)) + \frac{3p-1}{2} |g(x) - g(y)|^2 \leq K|x - y|^2.$$

Assumption 11 (Khasminskii)

There are constants $q > 2$ and $K_1 > 0$ such that, for all $x \in \mathbb{R}^d$,

$$x^\top f(x) + \frac{3q-1}{2} |g(x)|^2 \leq K_1(1 + |x|^2).$$

The TEM scheme

For super-linear coefficients, we truncate only the arguments of f and g .

Choose a strictly increasing envelope μ and a strictly decreasing cap κ satisfying

$$\sup_{|x| \leq u} (|f(x)| \vee |g(x)|) \leq \mu(u), \quad \mu(u) \rightarrow \infty,$$

$$\kappa(\Delta) \rightarrow \infty, \quad \Delta^{(\alpha-\varepsilon)/4} \kappa(\Delta) \leq \widehat{\kappa}.$$

Set

$$R_\Delta := \mu^{-1}(\kappa(\Delta)), \quad \pi_\Delta(x) := \left(1 \wedge \frac{R_\Delta}{|x|}\right) x, \quad \pi_\Delta(0) = 0,$$

$$f_\Delta := f \circ \pi_\Delta, \quad g_\Delta := g \circ \pi_\Delta, \quad |f_\Delta(x)| \vee |g_\Delta(x)| \leq \kappa(\Delta).$$

On $t_n = n\Delta$,

$$X_{t_{n+1}} = X_{t_n} + f_\Delta(X_{t_n})\Delta E_n + g_\Delta(X_{t_n})\Delta B_n.$$

The processes $\widetilde{X}$ and $\overline{X}$ are defined as in the EM case, with f, g replaced by f_Δ, g_Δ .

Key TEM estimates

Under Assumptions 9–11, the following estimates hold.

Lemma 12 (Li X., Liu W., Tang T., JCM, 2025)

$$x^\top f_\Delta(x) + \frac{3q-1}{2} |g_\Delta(x)|^2 \leq \tilde{K}_1(1 + |x|^2),$$

$$|f_\Delta(x) - f_\Delta(y)| \vee |g_\Delta(x) - g_\Delta(y)| \leq L(1 + |x|^\gamma + |y|^\gamma)|x - y|.$$

Lemma 13 (moment bounds)

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X(t)|^p \right] < \infty, \quad 2 \leq p < q,$$

$$\sup_{0 < \Delta \leq 1} \mathbb{E} \left[\sup_{0 \leq t \leq T} |\bar{X}(t)|^p \right] \leq C, \quad p \geq 2.$$

Lemma 14 (local interpolation error)

For $t \in [t_n, t_{n+1})$ and $\bar{p} \geq 2$, almost surely,

$$\mathbb{E}_B[|\bar{X}(t) - \tilde{X}(t)|^{\bar{p}}] \leq C\kappa(\Delta)^{\bar{p}}(\Xi_{\Delta}^{\bar{p}} + \Xi_{\Delta}^{\bar{p}/2}).$$

TEM convergence

Theorem 15 (truncated EM)

Let Assumptions 9–11 hold, let $\bar{p} \in [2, p)$ and $\varepsilon \in (0, \alpha)$. Suppose

$$q > (\gamma + 1)p, \quad \frac{\varepsilon(q - (\gamma + 1)\bar{p})}{4(\gamma + 2)} \geq \frac{\bar{p}}{2}(\alpha - \varepsilon).$$

Then

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X(t) - \bar{X}(t)|^{\bar{p}} \right] \leq C \Delta^{\frac{\bar{p}}{2}(\alpha - \varepsilon)},$$

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X(t) - \tilde{X}(t)|^{\bar{p}} \right] \leq C \Delta^{\frac{\bar{p}}{2}(\alpha - \varepsilon)}.$$

for all $\Delta \in (0, 1]$.

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Positive drift makes E Lipschitz

Positive drift let $D(t) = \vartheta t + \tilde{D}(t)$ with $\vartheta > 0$ and $\tilde{D}$ driftless. We also assume that the equation is globally well posed.

Lemma 16 (Zhang S., Chen Z.-Q., SICON, 2024)

Let D have drift $\vartheta > 0$. Then, almost surely, for all $0 \leq s < t$,

$$0 \leq E(t) - E(s) \leq \frac{t-s}{\vartheta}.$$

Thus E is Lipschitz with constant $1/\vartheta$.

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From exponential to polynomial decay

Regular variation at zero. We assume that the Laplace exponent ψ is regularly varying at zero with index $\alpha \in (0, 1)$:

$$\lim_{s \downarrow 0} \frac{\psi(cs)}{\psi(s)} = c^\alpha, \quad c > 0.$$

For example, $\psi(s) = \vartheta s + s^\alpha$ has index α .

Lemma 17 (Deng C., Liu W., BIT, 2020)

Let ψ be regularly varying at zero with index $\alpha \in (0, 1)$. Then, for each $\lambda > 0$,

$$\lim_{t \rightarrow \infty} \frac{\log \mathbb{E}_D[e^{-\lambda E(t)}]}{\log t} = -\alpha.$$

EM polynomial stability

Assumptions 18–19 (linear growth and dissipativity)

There are constants $K > 0$ and $L > 0$ such that, for all $x \in \mathbb{R}^d$,

$$|f(x)| \leq K|x|, \quad 2\langle x, f(x) \rangle + |g(x)|^2 \leq -L|x|^2.$$

Theorem 20

Under the positive-drift setting above, assume global well-posedness and Assumptions 18–19. Let $X_0 \in \mathbb{R}^d$ be deterministic, and let ψ be as in Lemma 17. Then there exists a sufficiently small $\Delta t^* \in (0, \min\{\frac{\vartheta}{L}, \frac{L\vartheta}{2K^2}\})$ such that, for every $\Delta t \in (0, \Delta t^*)$,

$$\limsup_{n \rightarrow \infty} \frac{\log \mathbb{E}[|X_n|^2]}{\log(n\Delta t)} \leq -\alpha.$$

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TEM asymptotic stability

We use the same TEM scheme and truncation functions defined in the convergence analysis.

Assumption 21 (nonlinear dissipation)

There is a continuous non-decreasing ρ with $\rho(0) = 0$ and $\rho(r) > 0$ for $r > 0$ such that, for all $x \in \mathbb{R}^d$,

$$2\langle x, f(x) \rangle + |g(x)|^2 \leq -\rho(|x|).$$

Theorem 22

Assume global well-posedness and Assumption 21. Let f and g be continuous with $f(0) = g(0) = 0$, and assume $\limsup_{|x| \rightarrow 0} |f(x)|^2 / \rho(|x|) < \infty$. Set

$$H := \sup_{0 < |x| \leq \mu^{-1}(\kappa(1))} \frac{|f(x)|^2}{\rho(|x|)}, \quad \Delta^* := \min \left\{ 1, \frac{\vartheta}{2H}, \left(\frac{\vartheta \rho(\mu^{-1}(\kappa(1)))}{2\hat{\kappa}^2} \right)^2 \right\}.$$

Then $H < \infty$ and, for every $\Delta \in (0, \Delta^*]$,

$$\lim_{n \rightarrow \infty} X_{t_n} = 0 \quad \text{a.s.}$$

Outline

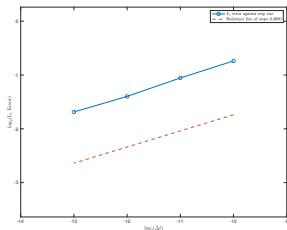
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Example 1: EM convergence

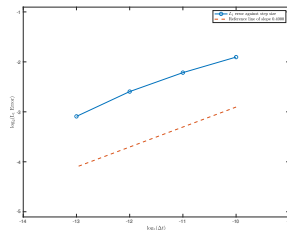
We first consider the following two-dimensional linear system.

$$\begin{cases} dX_1 = -(X_1 + X_2)dE(t) + 2(X_1 + X_2)dB_1(E(t)), \\ dX_2 = -2(X_1 + X_2)dE(t) + (X_1 + X_2)dB_2(E(t)), \end{cases} \quad X_1(0) = 1, \quad X_2(0) = 2.$$

We use the uniform L^1 error with $T = 1$, $M = 10^3$, and reference step 2^{-20} .



$\alpha = 0.6$



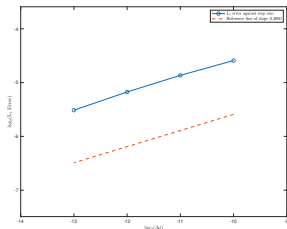
$\alpha = 0.8$

Example 2: TEM convergence

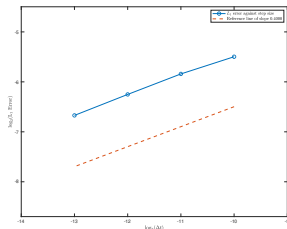
We next consider a two-dimensional system with super-linear coefficients.

$$\begin{cases} dX_1 = (X_1^2 - 2X_1^5)dE(t) + X_2^2dB_1(E(t)), \\ dX_2 = (X_2^2 - 2X_2^5)dE(t) + X_1^2dB_2(E(t)), \end{cases} \quad X_1(0) = 1, \quad X_2(0) = 2.$$

Use $\mu(u) = 2u^5$ and $\kappa(\Delta) = \Delta^{-0.01}$.



$\alpha = 0.6$



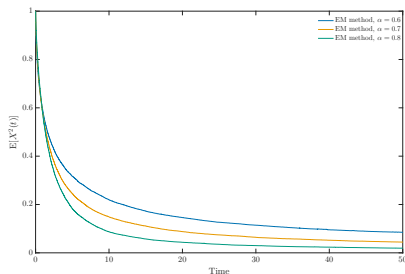
$\alpha = 0.8$

Example 3: EM polynomial stability

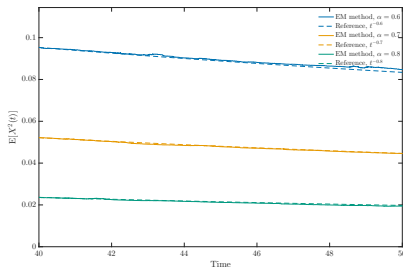
To illustrate mean-square polynomial stability, consider the scalar linear equation

$$dX(t) = -\frac{1}{3}X(t)dE(t) + \frac{1}{3}X(t)dB(E(t)), \quad X_0 = 1,$$

We set $\vartheta = 1$.



$E|X_n|^2$ for $\alpha \in \{0.6, 0.7, 0.8\}$



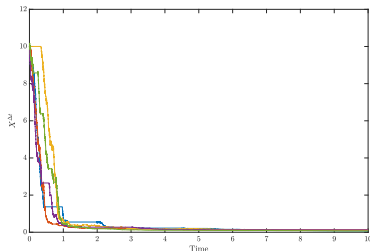
against $t^{-\alpha}$ on $[40, 50]$

Example 4: TEM asymptotic stability

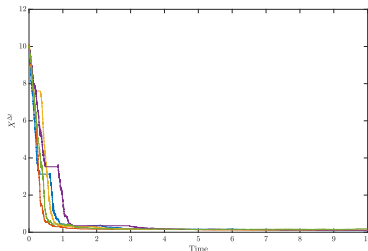
To illustrate the almost sure stability of TEM, consider the following super-linear equation.

$$dX(t) = -10X^3(t)dE(t) + X^2(t)dB(E(t)), \quad X_0 = 10.$$

Set $\vartheta = 1$, $\mu(u) = 10u^3$, $\kappa(\Delta) = 10\Delta^{-0.1}$ and $\Delta = 2^{-13}$.



$\alpha = 0.6$



$\alpha = 0.8$

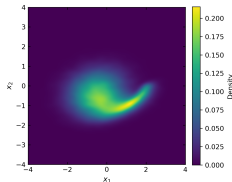
Example 5: FFPE comparison

Finally, consider the following two-dimensional linear system.

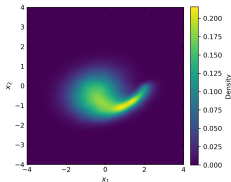
$$\begin{cases} dX_1(t) = (-0.8X_1(t) + 2X_2(t))dE(t) + dB_1(E(t)), \\ dX_2(t) = (-2X_1(t) - 0.8X_2(t))dE(t) + dB_2(E(t)), \end{cases} \quad X_1(0) = 2, \quad X_2(0) = 0.$$

The corresponding FFPE is

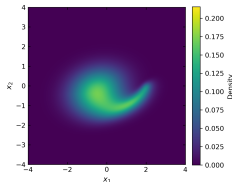
$$\begin{aligned} D_t^\alpha p &= \frac{\partial}{\partial x_1} [(0.8x_1 - 2x_2)p] + \frac{\partial}{\partial x_2} [(2x_1 + 0.8x_2)p] \\ &\quad + \frac{1}{2} \left(\frac{\partial^2 p}{\partial x_1^2} + \frac{\partial^2 p}{\partial x_2^2} \right), \quad p(x_1, x_2, 0) = \delta(x_1 - 2)\delta(x_2). \end{aligned}$$



(a) Equidistant physical-time grid



(b) Non-equidistant D -grid



(c) FFPE: L1 in time and finite

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- ② Z. Rong and Z. *Stability of equidistant-step Euler-type methods for time-changed stochastic differential equations. Submitted.*
- ③ R. Wu and Z. *A Milstein-type method for highly non-linear non-autonomous time-changed stochastic differential equations. Applied Numerical Mathematics*, 230: 209–226, 2026.

Thank you for your attention!

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